Coffee-Linked Structured Note

Structured Securities Proposal & Risk Mitigation Report | Scenario 2: Moderate-Risk Client

Executive recommendation

Recommend an illustrative six-month, cash-settled coffee-linked note combining a small fixed coupon, principal repayment at maturity and limited participation in coffee price increases above a defined strike. The design responds to the client's need for income and some commodity exposure without a direct position in futures.

This is a preliminary training-case structure, not a tradable quote. Principal repayment depends on the issuer's ability to pay; secondary-market prices can fall below par.

Proposed terms

Term	Illustrative specification
Investor profile	Moderate risk; seeks income and positive coffee participation
Initial investment / maturity	$100 note denomination / six months
Underlying / currency	Coffee reference price, USD per pound; cash settlement
Reference level / strike	Initial spot $1.20/lb; upside strike $1.25/lb
Fixed coupon	0.20% total over six months, paid at maturity
Upside participation	10% of coffee appreciation above the $1.25/lb strike
Principal protection	100% contractual principal at maturity, subject to issuer credit
Observation / settlement	One fixing at maturity; detailed market conventions to be set

Contractual payoff and investor outcomes

For each $100 invested, maturity payment = $100 × [1 + 0.002 + 0.10 × max((C_T − 1.25)/1.25, 0)]. Here C_T is the defined coffee reference fixing (USD/lb) at the six-month maturity.

Coffee at maturity	Investor payout / $100	Interpretation
$0.95/lb	$100.20	Coffee falls: coupon and principal only
$1.20/lb	$100.20	Unchanged versus initial spot: coupon only
$1.25/lb	$100.20	At strike: no upside participation
$1.40/lb	$101.40	12% rise above strike × 10% participation
$1.60/lb	$103.00	28% rise above strike × 10% participation

Contractual payments exclude fees and taxes. No issuer-default recovery is assumed in this table.

Quantitative valuation: models from the previous task

Reuse Task 2 inputs as a controlled example: spot $1.20/lb, risk-free rate 2% annually, storage carry 1% annually, volatility 25% annually, and six-month maturity. These are training assumptions, not current market quotations.

1 Cost-of-carry futures benchmark

$F_0 = S_0 \times \exp[(r + d) \times T] = 1.20 \times \exp[(0.02 + 0.01) \times 0.5] = \$1.21814$/lb.

The simple model assumes continuously compounded financing/storage rates and zero convenience yield. Before trading, calibrate delivery grade, location, inventories, convenience yield, forward curve and any spot-to-futures basis.

2 Option component and indicative funding budget

For a European payoff on a futures-related coffee reference, use a Black-76-style formula: $\text{Call} = \exp(-rT)[F_0 N(d1) - K N(d2)]$, with $d1 = [\ln(F_0/K) + 0.5\sigma^2T]/(\sigma\sqrt{T})$, $d2 = d1 - \sigma\sqrt{T}$, and $K = \$1.25$/lb.

The theoretical call value is $0.07119/lb. The note embeds 10% participation in $(C_T - K)/K$, equivalent to a call budget of $100 × 0.10/K × $0.07119 = $0.5695 per $100 of principal.

Indicative cost at issue / $100	Present value
Risk-free funding of principal at maturity	$99.0050
PV of $0.20 coupon at maturity	$0.1980
PV of embedded coffee call participation	$0.5695
Total theoretical cost	$99.7725
Unallocated headroom before all costs	$0.2275

Headroom is only an indicative model budget, not an issuer margin: transaction fees, market-making hedges, issuer funding/credit, model reserves and slippage must be priced before issuance.

3 Monte Carlo validation and scenario analysis

Simulate 200,000 independent six-month terminal coffee prices with geometric Brownian motion under a risk-neutral drift matching the simplified forward: $C_T = S_0 \times \exp[(r + d - 0.5\sigma^2)T + \sigma\sqrt{T} \times Z]$, $Z \sim N(0,1)$; random seed 42.

Monte Carlo call estimate = $0.07141/lb (95% simulation confidence interval $0.07084–$0.07197/lb), consistent with the analytical $0.07119/lb value. This is a model consistency check, not an empirical forecast.

For supply-demand, climate and geopolitical stresses, perturb the forward curve and volatility separately. Strong harvests may lower expected prices; crop damage, export restrictions and logistics shocks may raise prices and volatility. No event probabilities are implied by these directional scenarios.

Hedging implementation and risk mitigation

Issuer hedge: buy a matching coffee call in the market where available, or dynamically delta-hedge the short embedded call exposure with the appropriate futures contract. Fund the protected payment with high-quality discounted assets matched to maturity, subject to funding and counterparty limits.

Risk	Mitigation and control
Coffee price / delta	Daily option Greeks; set delta/vega limits, rebalance hedge and stress large gaps.
Basis, delivery and roll	Match ICE grade, reference fixing, expiry and units; monitor spot–futures basis and roll.
Volatility / model risk	Compare implied and historical volatility; reprice with Black-76 and Monte Carlo; independent validation.
Liquidity / execution	Limit position size, maintain liquid hedging instruments and assess bid-ask spreads.
Issuer / counterparty credit	Disclose that protection is contractual, not insured; set exposure limits and review credit quality.
Financing / interest rates	Match funded liabilities and maturities; monitor discounting and funding spread changes.
Operations / legal	Daily reconciliation; confirm documentation, fixings, settlement and investor disclosures.

Key sensitivity: volatility can exhaust pricing headroom

Implied volatility	Call fair value	Note option cost / $100	Theoretical headroom
15%	$0.03742/lb	$0.2994	$+0.4976
25%	$0.07119/lb	$0.5695	$+0.2275
35%	$0.10520/lb	$0.8416	$-0.0446

At 35% volatility, the structure becomes theoretically underfunded at the same offered terms. Reprice participation/coupon, reduce exposure, obtain executable hedges or decline issuance; never assume a historical-volatility estimate is an executable option quote.

Decision and next steps

Proceed only to an indicative client discussion. Before issuance: (i) obtain the relevant coffee futures and option quotes, (ii) confirm legal reference/settlement terms and suitability, (iii) validate funding and hedge costs, (iv) run downside, gap, volatility and default stresses, and (v) secure independent model and risk approval.

Provenance: client scenarios and product techniques are supplied by Citi/Forage Task 3; price inputs and core models are reused from the earlier Task 2 learning exercise. The coupon, strike use, participation rate, structure, risk limits and Monte Carlo valuation here are analyst-designed illustrative assumptions, not statements about Citi products.